

README: An Enhanced Spine Model Validated for Simulating Dynamic Lifting Tasks in OpenSim

Introduction

Welcome to the repository for our spine model developed in OpenSim 3.3. The intervertebral spinal forces estimated by this model using five external hand forces and moments (EHF&M) modeling approaches were validated. In this README file, we provide detailed instructions on how to use our model files, setup files, and MATLAB scripts for your research purposes.

Important References

- For a comprehensive understanding of our model and methodology, please refer to our paper:

Akhavanfar, M., Mir-Orefice, A., Uchida, T. k., & Graham, R. B. (2023). An Enhanced Spine Model Validated for Simulating Dynamic Lifting Tasks in OpenSim. *Annals of Biomedical Engineering*. <https://doi.org/10.1007/s10439-023-03368-x>

- Detailed information about the EHF&M approaches can be found in:

Akhavanfar, M., Uchida, T. K., Clouthier, A. L., & Graham, R. B. (2022). Sharing the load: modeling loads in OpenSim to simulate two-handed lifting. *Multibody System Dynamics*, 54(2), 213–234. <https://doi.org/10.1007/s11044-021-09808-7>

Contents

This repository includes the following:

1. **Spine Model:** Our newly developed spine model in OpenSim 3.3.
2. **Sample Setup Files:** These files are essential for obtaining muscle forces and joint loads in OpenSim.
3. **MATLAB Scripts:** These scripts automatically generate the models and motion files needed for the kinematic and dynamic analysis of EHF&M Approaches 3–5 using the files required for Approaches 1 and 2 as a starting point.

Please note that the model was created in OpenSim 3.3, and as a result, the scripts are designed to use the API for OpenSim 3.3. However, the model can be used in more recent versions, such as OpenSim 4.4. To run the MATLAB scripts successfully, ensure that you have the OpenSim 3.3–MATLAB environment set up and uninstall other versions. For setup instructions, please refer to:

<https://simtk-confluence.stanford.edu:8443/display/OpenSim33/Scripting+with+Matlab>.

The MATLAB scripts require access to the OpenSim libraries to call OpenSim methods.

Participant and Task Selection

We evaluated our model using data from 3 participants (P1, P2, and P3) and during 9 dynamic lifting tasks (Akhavanfar et al., 2023). In total, we conducted 531 dynamic simulations in our recent study. To simplify usage and avoid confusion, we are providing the main files and scripts for one participant (P2) and one task (Squat-7). You can modify these models and files for your specific needs.

- **Participant Selection:** We arbitrarily selected P2 as the example participant. You may scale this model to match your subjects.
- **Task Selection:** Squat-7 was chosen as an example because setting up the model for Approaches 3-5 is more challenging in two-handed lifting tasks, as explained in (Akhavanfar et al., 2022). However, the provided scripts can be adapted for other participants and tasks using *simple for loops* in MATLAB.

Directory Structure

For reference, the directory structure is as follows:

- **Squat-7 Folder:** Contains all required files for running Inverse Kinematics (IK), Static Optimization (SO), and Joint Reaction (JR) analyses in OpenSim.
- **Squat-7-withOutputs Folder:** For your reference, this folder contains all inputs and outputs generated after running all MATLAB codes and setup files.

Getting Started

Please follow the step-by-step instructions in this README file to adequately set up and run IK, SO, and JR analysis in OpenSim. After executing the analyses, the outputs will appear in the respective subdirectories within the Squat-7 folder.

1. Squat-7 Folder

The Squat-7 folder contains essential files and folders for setting up and running the spine model using the EHF&M modeling approaches (APP1-5). Here's an overview of its contents:

1.1. Files:

- **Squat7kgBothHandsCyl1.trc:** This file contains passive marker data collected using an 11 infrared camera system (Vantage V5, Vicon) for the purposes of this study.
- **Squat_7kg_BothHands_Cyl1.mot:** This file contains force plate data also collected for the purposes of this study and exported using Vicon Nexus.
- **7kgBox_MrkGND_Sample.osim:** This file represents a sample model of a 7 kg box with marker locations on ground (GND) reference frame. The marker locations in this model

do not correspond to the Squat-7 task performed by P2. This sample file will be used as a basis to create the box model for the specific trial (explained later).

1.2. MATLAB Codes Folder:

The “Matlab Codes” folder is organized into subdirectories and contains all the MATLAB codes necessary for generating specific files related to EHF&M approaches 3–5. Here’s a breakdown of its contents:

- **APP3:** This subdirectory contains MATLAB scripts for EHF&M Approach 3.
- **APP4:** This subdirectory contains MATLAB scripts for EHF&M Approach 4.
- **APP5:** This subdirectory contains MATLAB scripts for EHF&M Approach 5.
- **General:** This subdirectory contains general-purpose MATLAB scripts that are useful across all EHF&M approaches and are not specific to a certain approach.

2. Squat-7>Matlab Codes>General

2.1. Function Installation:

- Locate “OriCalc.m”, “RotMat.m”, “normalize.m”, and “NonlinearConst.m” functions in the “General” folder.
- Copy these functions to your MATLAB installation directory as these functions might be called within other codes.

2.2. Generating the Box Model:

- Run “LoadModelGenMrkGnd.m” followed by “LoadModelGen.m” in the “General” folder.
 - These scripts create two model files in the “Squat-7” folder: “Load_MrkGND.osim” and “Load.osim.”
 - These models represent the box with 8 markers for the specific trial in both ground and box reference frames.
- Copy both model files into “Squat-7>APP3”.
- Copy both model files into “Squat-7>APP5”.

3. Squat-7>APP1

3.1. Inverse Kinematics (IK)

- Locate the spine model developed in this study, named “P2_APP1_IK.osim”, within the folder.

- Open this model in OpenSim 3.3.
- In the Inverse Kinematics Tool, load the setup file “IKSetup.xml”.
- Execute the IK analysis and save the results as “IKResults.mot” in this folder.

3.2. Body Kinematics (BK) Analysis

- Open the “P2_APP1_SO.osim” model, which is identical to “P2_APP1_IK.osim” but without kinematic constraints (as explained in Akhavanfar et al., 2023).
- Open the Analyze Tool in OpenSim and load the setup file “BKSetup.xml”.
- Run the Analyze Tool and the resulting BK data for hands will be saved in the “Squat-7>APP1>BK” folder.
- In the “Squat-7>APP1” folder, you will find the “motioncycletimes.m” file, which contains a 1×4 matrix indicating these key instants: i) start of the lifting, ii) end of lifting/start of holding, iii) end of holding/start of lowering, and iv) end of lowering.

Note that performing BK analysis for hands are required to determine motion cycles (see Akhavanfar et al., 2023). BK results for hands are also required for generating appropriate models in Approaches 3 and 4. Thus, researchers planning to utilize only Approach 3 or 4 for estimating muscle forces and joint loads should perform this BK analysis as a prerequisite.

3.3. Static Optimization (SO)

To estimate muscle forces, open the “P2_APP1_SO.osim” model and follow these steps:

- Open the SO Tool in OpenSim.
- Load the setup file “SOSetup.xml”.
- Running SO requires the following additional files, which are already provided in the “Squat-7>APP1” folder:
 - “IKResults.mot” from the IK analysis performed in step 3.1.
 - “ExtLoad.xml” (external load setup).
 - “P2Reserves.xml” (reserve actuators setup).
- The “ExtLoad.xml” file applies external forces to the model based on the “ExtForce.mot” file. “ExtForce.mot” contains measured ground reaction forces as well as external forces applied to the hands. Details regarding forces applied to the hands can be found in (Akhavanfar et al., 2022). In brief, hand loads in Approach 1 are modeled as constant gravity-oriented loads (i.e., 34.3 N to each hand in this trial) between the start of lifting and the end of lowering. Hand load locations are the positions of hand markers, which are extracted from the measured marker data.

- After running SO, you will find muscle activations and forces in the “SO” subdirectory.

3.4. Joint Reaction (JR) Analysis

To obtain joint reaction forces, use the following steps for the “P2_APP1_SO.osim” model:

- Open the Analyze Tool in OpenSim.
- Load the configuration file “JRSetup.xml”.
- The computed joint reaction forces will appear in the “JR” subdirectory.

4. Squat-7>APP2

4.1. IK

- The IK analysis for Approach 2 is the same as that for Approach 1, thus copy “IKResults.mot” from the “Squat-7>APP1” folder into this folder.

4.2. SO and JR Analysis

- Use the “P2_APP2_SO.osim” model for SO and JR analyses in this approach. This model is similar to “P2_APP1_SO.osim” with the only difference being that each hand mass has been increased by half of the lifted mass (i.e., 3.5 kg in this trial).
- The provided “ExtForce.mot” is also similar to the one provided for APP1, with the key difference being that, in this file, forces applied to the hands are set to zero since these forces were modeled as the increased mass of the hands.
- Open the Static Optimization (SO) Tool in OpenSim.
- Load the setup file “SOSetup.xml”.
- After running SO, you will find muscle activations and forces in the “SO” subdirectory.
- The steps of running JR analysis are the same as those outlined above in Section 3.4 for APP1.

5. Squat-7>APP3

5.1. IK

- To generate the necessary IK analysis model for Approach 3, run the “IKModelGenAPP3.m” MATLAB script located in the “Squat-7>Matlab Codes>APP3” folder. Please read the following points in Section 5.1 to ensure successful execution of this script. The “IKModelGenAPP3.m” generates three models in the directory, namely, “FlexInit.osim”, “FlexInitWithBox.osim”, and “P2_APP3_IK.osim”.
 - “FlexInit.osim” is generated at the start of the lifting phase by setting the default coordinate values of the original model (“P2_APP1_IK.osim”) to match the values

from “IKResults.mot” at that specific moment. Before running this script, ensure that “P2_APP1_IK.osim” and “IKResults.mot” from the “Squat7>APP1” folder are either copied to the current directory or their file locations in the MATLAB script are adjusted accordingly.

- “FlexInitWithBox.osim” is a model where each hand is connected to one half-box through a joint, and a weld constraint is created between the two half-boxes. The model assembly complexity is detailed in Akhavanfar et al., (2022). To create this model, the MATLAB script requires the following files: i) “Load.osim” (explained in section 2.2), ii) collected marker data (i.e., “Squat7kgBothHandsCyl1.trc”), iii) BK data for hands (explained in section 3.2), and iv) “FlexInit.osim” generated during code execution.
- The “P2_APP3_IK.osim” model, intended for IK analysis, is identical to “FlexInitWithBox.osim,” but it includes the box markers required for the analysis.
- Open “P2_APP3_IK.osim” in OpenSim.
- In the Inverse Kinematics Tool, load the setup file “IKSetupAPP3.xml”.
- Execute the IK analysis and save the results as “IKResultsAPP3.mot” in this folder.

5.2. SO and JR Analysis

- First, run the “SOModelGenAPP3.m” MATLAB script located in the “Squat-7>Matlab Codes>APP3” folder. This script generates “P2_APP3_SO.osim” which will be used for both SO and JR analysis. The script removes all kinematic constraints from the “P2_APP3_IK.osim” model, except for the Weld constraint between the half-boxes.
- As detailed in Akhavanfar et al., (2022), Approach 3 requires special consideration when running SO due to acceleration-level constraint violations arising from the nonlinear algebraic constraint equations added in the closed-kinematic-chain systems. While for other approaches, coordinate angles (from IK results) can be filtered using the filter coordinates option in the SO Analysis Tool, using the same method in Approach 3 can lead to assembly errors in the model. Conversely, not filtering coordinates can result in high constraint violations due to acceleration noise. To avoid assembly errors and mitigate constraint violations in Approach 3, an alternative method is to leave the coordinates unfiltered but filter the coordinate velocities. To implement this, you’ll need to use the States Analysis feature in OpenSim, which encompasses both coordinate angles and velocities. Therefore, before conducting SO analysis, follow these steps:
 - Open the “P2_APP3_SO.osim” model.
 - Execute the Analyze Tool twice: once with the “StatesReporter_Filt_Setup.xml” setup file and once with the “StatesReporter_NoFilt_Setup.xml” setup file. The

outcomes of these analyses will be stored in the “Squat7>APP3>States” folder as “Filt_StatesReporter_states.sto” and “NoFilt_StatesReporter_states.sto”, respectively.

- Copy the file “NoFilt_StatesReporter_states.sto” to the Squat7>APP3 folder and rename it as “Combined_StatesReporter_states.sto”. In the states file, coordinate angles are reported first, followed by coordinate velocities. Their names follow the format of “Coordinate Name_u.” For example, the first coordinate in the model is “pelvis_tilt”, and its corresponding velocity column is labeled “pelvis_tilt_u”. To complete the task, copy and replace the columns after the headers “pelvis_tilt_u” from “Filt_StatesReporter_states.sto” into “Combined_StatesReporter_states.sto” and save it.
- External forces setup in Approaches 3 and 4 is identical to Approach 2. Therefore, copy the file “ExtForce.mot” from the “Squat-7>APP2” folder to “Squat-7>APP3” and “Squat-7>APP4” folders.
- With all the necessary files prepared, open SO and Analyze Tool in OpenSim. Run these analyses for the “P2_APP3_SO.osim” model in OpenSim using the respective setup files “SOSetup.xml” and “JRSetup.xml”.

6. Squat-7>APP4

6.1. IK

IK analysis in Approach 4 is identical to Approach 3. Therefore, copy the file “IKResultsAPP3.mot” from “Squat-7>APP3” to “Squat-7>APP4”.

6.2. SO and JR Analysis

- Run the “SOModelGenAPP4.m” MATLAB script located in the “Squat-7>Matlab Codes>APP4” folder. This script removes the Weld Constraint between the two half-boxes from “P2_APP3_SO.osim”, creating “P2_APP4_SO.osim” for SO and JR analyses.
- As mentioned in Section 5.2, the external forces setup for Approach 4 is identical to Approach 2. If not done already, copy the file “ExtForce.mot” from “Squat-7>APP2” to “Squat-7>APP4”.
- With the Weld Constraint removed, the model adopts an open tree structure similar to Approaches 1 and 2, allowing SO and JR analyses to be performed with filtered IK results. Execute these analyses for the “P2_APP4_SO.osim” model in OpenSim using the setup files “SOSetup.xml” and “JRSetup.xml”.

7. Squat-7>APP5

7.1. IK

In this approach, the kinematics of the lifted mass is calculated separately from the body kinematics. Follow these steps:

- Run the “LoadIKAPP5.m” MATLAB script located in the “Squat-7>Matlab Codes>APP5” folder. This script is responsible for conducting the IK analysis for the “Load.osim” model. If you have not yet generated the “Load.osim” model and copied it into the “Squat-7>APP5” folder, please refer to section 2.2 above. Executing “LoadIKAPP5.m” will generate the “IKLoadSetup.xml” and “IKResultsLoad.mot” files in the “Squat-7>APP5” directory.
- The human body kinematics should be those calculated using Approach 1. Therefore, copy the “IKResults.mot” file from the “Squat-7>APP1” folder to the “Squat-7>APP5” folder.
- Finally, execute the “CombIKResultsAPP5.m” MATLAB script located in the “Squat-7>Matlab Codes >APP5” folder. This script combines the data from “IKResultsLoad.mot” and “IKResults.mot” to create the “IKResultsCombined.mot” file.

7.2. Load BK and States Reporter Analysis

To calculate external forces applied to the hands in Approach 5, the translational and rotational displacements, velocities, and accelerations of the lifted mass are required (Akhavanfar et al., 2022). Therefore, please run the “LoadBKAPP5.m” and “LoadStatesAPP5.m” MATLAB scripts located in the “Squat-7>Matlab Codes>APP5” directory. These scripts will generate BK and state results for the load, which can be found in the “Squat-7>APP5>LoadAnalysis” directory.

7.3. SO and JR Analysis

- The body model in Approach 5 is the same as that in Approach 1. Therefore, copy “P2_APP1_SO.osim” from “Squat-7>APP1” to “Squat-7>APP5” directory. Use the “CombOsimModelsAPP5.m” MATLAB script located in the “Squat-7>Matlab Codes>APP5” folder to combine “P2_APP1_SO.osim” and “Load.osim” models into the model that should be used for SO and JR analysis in Approach 5, which is “P2_APP5_SO.osim”.
- To generate the “ExtForceAPP5.mot” file, which serves as the external forces file for SO and JR analysis in Approach 5, execute the “ExtForceGenAPP5.m” MATLAB script. This script can be found in the “Squat-7>Matlab Codes>APP5” directory. Note that this code reads the force plate data from the external forces file prepared for Approach 1. Therefore, copy the “ExtForce.mot” file from the “Squat-7>APP1” directory to the

“Squat-7>APP5” directory, and rename it to “ExtForceAPP1.mot” to avoid confusion with the “ExtForceAPP5.mot” file generated by the “ExtForceGenAPP5.m” script. The “ExtForceGenAPP5.m” MATLAB script utilizes BK and States results (explained in section 7.2) to calculate the forces applied to the right and left hands and their corresponding points of application at the center of pressure for each hand. For a more detailed explanation, please refer to Akhavanfar et al. (2022).

- Before running SO and JR analysis, it is important to note the differences in the external load and reserve actuator setup files in Approach 5 compared to other approaches, as understanding these distinctions will assist users who intend to modify and utilize these setup files for their specific applications.
 - The external load setup file in Approach 5, named “ExtLoadAPP5.xml”, differs from that in other approaches (i.e., “ExtLoad.xml”). While all external load setup files specify that hand forces are applied to the right and left hands, there are two significant distinctions: 1) In “ExtLoad.xml”, the reference frame in which hand forces are expressed is ground, while in “ExtLoadAPP5.xml”, it is the load coordinates. 2) Similarly, in “ExtLoad.xml” the reference frame in which the point of application of hand forces is expressed is ground, while in “ExtLoadAPP5.xml”, it is the load reference frame.
 - The reserve actuator setup files in Approach 5, named “P2ReservesAPP5.xml” also differ slightly from those in other approaches (“P2Reserves.xml”). This difference arises because in the combined model of Approach 5 (“P2_APP5_SO.osim”), the box body is attached to the ground, necessitating reserve actuators for its coordinates. Therefore, in addition to the actuators found in “P2Reserves.xml”, “P2ReservesAPP5.xml” includes 6 coordinate actuators corresponding to the 6 coordinates defined between the box and the ground.
- Execute SO and JR analysis for the “P2_APP5_SO.osim” model using the setup files “SOSetup.xml” and “JRSetup.xml”.